

Kushagra Bhargava

+91 9779070316 | kushagra2112004@gmail.com | linkedin.com/in/kushagra-bhargava02 | github.com/KushagraBhargava26

EDUCATION

VIT Bhopal University

Bachelor of Technology in Computer Science & Engineering (AI & ML)

– Current CGPA: 8.32

Sehore, MP

Sep. 2023 – Apr. 2027

EXPERIENCE

Pinnacle Labs

C++ Developer Intern

Remote

Nov. 2024 – Dec. 2024

- **Engineered** a high-performance command-line number conversion tool in C++ utilizing **OOP** principles to ensure modularity and scalability.
- **Optimized** conversion algorithms for Binary, Octal, Decimal, and Hexadecimal systems, achieving **100% mathematical accuracy** across 500+ test cases.
- **Streamlined** development workflows by collaborating via **Git**, participating in peer code reviews that improved codebase maintainability by 15%.

PROJECTS

VyomLens (Geospatial Intelligence Engine) | PyTorch, Node.js, Express, Vanilla JavaScript, CSS

- **Architected** a cross-language application bridging a **Node.js/Express.js** backend with a headless **Python** inference engine via inter-process communication (IPC).
- **Trained** a **ResNet18 CNN** using **PyTorch** and transfer learning to autonomously classify high-resolution satellite imagery into 10 distinct topographical categories, achieving a **97.6% validation accuracy**.
- **Deployed** the full-stack environment seamlessly on Hugging Face Spaces, complete with a custom, interactive **Vanilla JavaScript** dashboard.

Plant Disease Classifier | Python, TensorFlow, Keras, MobileNetV2, Matplotlib

- **Architected** a deep learning model using **Transfer Learning (MobileNetV2)** to classify 38 plant diseases, achieving a **94% validation accuracy**.
- **Developed** a Python-based CLI tool to process leaf images, delivering real-time predictions with **90%+ confidence scores** for agricultural diagnostics.
- **Automated** dataset management and pipeline setup using **KaggleHub**, reducing data preprocessing time by 20%.

Mind Mate (AI-Powered Mental Health Companion) | XGBoost, NLP, React, Node.js

- **Built** an **XGBoost** predictive model achieving **78% accuracy** on test data to identify mental health needs and provide personalized coping strategies.
- **Deployed** a full-stack platform using the **MERN stack**, ensuring a seamless user experience for real-time mental health tracking.
- **Implemented** an AI chatbot with **Natural Language Processing (NLP)** to analyze user emotions and provide empathetic, real-time support.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Java, SQL, HTML/CSS

Web Development: React, Next.js, Node.js, Express.js, PostgreSQL, MongoDB, REST APIs

Developer Tools: Git, GitHub, Postman, Google Cloud Platform, VS Code

Libraries: PyTorch, Scikit-learn, XGBoost, Pandas, Matplotlib, NumPy, TensorFlow, Keras

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, System Design

CERTIFICATIONS

Google Cloud Generative AI Virtual Internship | SmartInternz

Jan. 2026

Google IT Support Professional Certificate | Google

Jan. 2026

C++ Programming with DSA | GeeksForGeeks

Feb. 2025